PLC Point List

Universal Plastic Shredder V2.0

The complete as-built PLC point list, transcribed sheet-by-sheet from `PLC/src/Point_List.xlsx`. Every sheet of the workbook is reproduced on the landscape pages below using the spreadsheet's columns: *Point Description*, *Origin Address*, *DO / DI / AO / AI / Pwr*, *Destination Address*, *Destination Description*, and *Notes*.

Sheet: Index

POINT ASSIGNMENT INDEX					
	PAGE NO.	NO. OF SHEETS	DESCRIPTION		
	1	1	Mitsubishi SD-N35		
	2	2	Mean Well NDR-480-24		
	4	1	Direct Logic 205 D2-09B-1		
	5	1	DirectLOGIC CPU (H2-DM1E)		
	6	1	Output Module (D2-08TA)		
	7	1	HMI (EA1-T4CL)		
	8	1	Output Module (F2-02DAS-2)		
	9	1	Input Module (D2-08ND3)		
	10	1	Analog Input (F2-08AD-1)		
	11	1	RTD Input (F2-04RTD)		
	12	1	Relay Output (D2-08TR)		
	13	1	Analog Output (F2-08DA-2)		
	14	3	VFD (Huanyang HY02D211B-T)		

Sheet: Index — Revision History

	REV	DATE	DESCRIPTION	Created By	Checked by
	1	2026-03-14	Initial point list (end of Winter term).	Bao Nguyen	Fearghus Tyler
	2	2026-05-18	Updated with as-built wiring and module corrections.	Bao Nguyen	Fearghus Tyler
	3	2026-06-01	Added missing component sheets to index; full revision log.	Bao Nguyen	Fearghus Tyler

Sheet: Mitsubishi SD-N35

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
Contactor (Mitsubishi SD-N35) – Coil A1	A1	0	0	0	0	1	Y0	D2-08TD2 Output Module (12–24VDC)	A1 → DF103V (1/2A slow blow fuse) → D2-08TD2 output supplying +24VDC
Contactor (Mitsubishi SD-N35) – Coil A2	A2	0	0	0	0	1	-V	NDR-480-24 Power Supply	A2 → direct return to power supply negative (–V)
Contactor (Mitsubishi SD-N35) – Line Input	L1	0	0	0	0	1	L	Phoenix Contact TMC 83C 15A (2907618)	L1 → Phoenix Contact fuse breaker → fed from wall AC line
Contactor (Mitsubishi SD-N35) – Load Output	T1	0	0	0	0	1	R	HY Series VFD (HY02D211B-T)	T1 → passes through contactor → supplies VFD terminal R (line input)
	Total Points	0	0	0	0	4			

Sheet: Mean Well NDR-480-24

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
		0					AC Mains Hot from wall via main power switch and 25 A fuse	120 VAC line conductor from wall outlet, routed through panel main disconnect, then through 25 A slow-blow fuse on EURO S10H-5H fuse block	Wall AC L -> main disconnect -> EURO S10H-5H 25 A slow-blow -> PSU L. Black wire per NFPA 79.
Mean Well NDR-480-24 N (AC Neutral Input)	N	0	0	0	0	1		120 VAC neutral conductor from wall outlet, returns through neutral bus	Wall AC N -> neutral bus -> PSU N. White wire per NFPA 79.
Mean Well NDR-480-24 G (Ground / PE)	G	0	0	0	0	1	Panel Ground Bus	Equipment grounding conductor; bonds PSU chassis to panel ground bus and ultimately to building earth	PSU G stud -> panel ground bus. Green or Green/Yellow wire per NFPA 79.
Mean Well NDR-480-24 +V to EURO S4LH (Pushbutton Common)	+V	0	0	0	0	1	EURO S4LH Terminal Block -> Pushbutton Commons	Fused +24V distribution to operator pushbutton common via S4LH	+V -> 1 A fast-blow fuse (EURO S4LH terminal) -> distributes to common side of all illuminated pushbuttons (Forward S2PR-P3G, Reverse S2PR-P3Y, Run 1/2 S2PR-P3B x2, Reset S2PR-P3R).
Mean Well NDR-480-24 +V to E22B1 E-Stop Circuit	+V	0	0	0	0	1	Cutler-Hammer E22B1 NC contact + SD-N35 contactor coil A1	Direct +24V (no fuse) into E-stop safety chain	+V -> E22B1 NC contact -> SD-N35 A1 (contactor coil). PLC X4 taps this line between E22B1 and A1 to monitor E-stop state.
Mean Well NDR-480-24 +V to VFD FA (Jam Signal)	+V	0	0	0	0	1	VFD Huanyang HY02D211B-T FA terminal	Direct +24V (no fuse) into VFD jam signal circuit	+V -> VFD FA. When VFD fault relay closes (PD052=02), FA-FC bridges and delivers +24V to PLC X5.
Mean Well NDR-480-24 +V to F2-08AD-1 (Module Power)	+V	0	0	0	0	1	F2-08AD-1 Analog Input Module +24V terminal	Direct module power	+V -> F2-08AD-1 +24V terminal. No fuse on this branch.
Mean Well NDR-480-24 +V to F2-08DA-2 (Module Power)	+V	0	0	0	0	1	F2-08DA-2 Analog Output Module +24V terminal	Direct module power	+V -> F2-08DA-2 +24V terminal. No fuse on this branch.
Mean Well NDR-480-24 +V to F2-04RTD (Module Power)	+V	0	0	0	0	1	F2-04RTD RTD Input Module +24V terminal	Direct module power	+V -> F2-04RTD +24V terminal. No fuse on this branch.
Mean Well NDR-480-24 +V to Indicator LEDs	+V	0	0	0	0	1	LED indicator (+) terminals (SA-LDG/Y/B/B/R, Wamco 525)	Direct +24V to (+) side of all illuminated pushbutton LED blocks and the separate Wamco error light	+V -> SA-LDG (+), SA-LDY (+), SA-LDB (+) x2, SA-LDR (+) (when wired), Wamco 525 (+). LED (-) sides return to D2-08TR relay outputs Y0-Y4 which sink to -V when energized.

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
Mean Well NDR-480-24 -V to SD-N35 Contactor Coil A2	-V	0	0	0	0	1	Mitsubishi SD-N35 Contactor A2 (coil return)	Contactor coil return through E-stop chain	-V -> SD-N35 A2 terminal. Coil energizes when E22B1 NC closes (E-stop not pressed) and +V reaches A1.
Mean Well NDR-480-24 -V to D2-08ND3 (Input Common)	-V	0	0	0	0	1	D2-08ND3 Discrete Input Module Common (C)	Module input common for sourcing mode operation	-V -> D2-08ND3 C terminal. Sourcing mode reference - X inputs read HIGH when +24V is applied.
Mean Well NDR-480-24 -V to D2-08TR (Relay Common)	-V	0	0	0	0	1	D2-08TR Relay Output Module Common (C)	Relay common for sinking mode operation	-V -> D2-08TR C terminal. When any Y0-Y7 relay closes, that output terminal connects to -V (sinks current from the load).
Mean Well NDR-480-24 -V to F2-08AD-1 (Module Power Return)	-V	0	0	0	0	1	F2-08AD-1 Analog Input Module 0V terminal	Module power return and analog signal common	-V -> F2-08AD-1 0V terminal.
Mean Well NDR-480-24 -V to F2-08DA-2 (Module Power Return / Analog Common)	-V	0	0	0	0	1	F2-08DA-2 Analog Output Module 0V terminal	Module power return and analog signal reference for all 8 channels	-V -> F2-08DA-2 0V terminal. Single 0V terminal also serves as the analog signal common.
Mean Well NDR-480-24 -V to F2-04RTD (Module Power Return)	-V	0	0	0	0	1	F2-04RTD RTD Input Module 0V terminal	Module power return	-V -> F2-04RTD 0V terminal.
Mean Well NDR-480-24 -V to VFD DCM (Digital Common)	-V	0	0	0	0	1	VFD Huanyang HY02D211B-T DCM terminal	VFD digital input common reference	-V -> VFD DCM. Provides 0V reference for FOR/REV/RST inputs (sinking/NPN mode). Closes the loop when D2-08TR relays Y5/Y6/Y7 pull those VFD inputs to DCM.
Mean Well NDR-480-24 -V to VFD ACM (Analog Common)	-V	0	0	0	0	1	VFD Huanyang HY02D211B-T ACM terminal	VFD analog signal common reference	-V -> VFD ACM. Shared 0V reference for the analog speed command (VI from F2-08DA-2) and the analog current monitor (VO to F2-08AD-1).
Mean Well NDR-480-24 +V to HMI (EA1-T4CL)	+V	0	0	0	0	1	C-More EA1-T4CL HMI +VDC input	HMI panel power	+V -> HMI +VDC. 24 VDC supply for the C-More Micro-Graphic panel (replaces former Rhino PSB12-030-P).
Mean Well NDR-480-24 -V to HMI (EA1-T4CL)	-V	0	0	0	0	1	C-More EA1-T4CL HMI 0VDC return	HMI panel power return	-V -> HMI 0VDC. System DC common reference for the HMI panel.
Total Points		0							

Sheet: Direct Logic 205 D2-09B-1

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
PLC Rack (DirectLOGIC 205) – Base	Base	0	0	0	0	1	D2-09B-1	Base / Power Unit	Rack base unit providing backplane power and I/O bus to all installed modules. Supports up to 9 I/O slots.
PLC Rack (DirectLOGIC 205) – Line Input	L	0	0	0	0	1	L	MORSETTITALIA EURO S10H-5H Fuse Block	Line from wall -> passes through power switch (closed = line passes) -> enters EURO S10H-5H fuse block with 1A fast blow fuse (PLC branch) -> enters PLC L terminal
PLC Rack (DirectLOGIC 205) – Neutral Input	N	0	0	0	0	1	N	Neutral Bus	Neutral from wall -> directly connected to PLC N terminal
PLC Rack (DirectLOGIC 205) – Ground (Protective Earth)	G	0	0	0	0	1	G	Ground Bus	Protective earth -> connected to PLC ground terminal / chassis bond
PLC Rack (DirectLOGIC 205) – Slot 1	Slot 1	0	0	0	0	0	H2-DM1E	CPU / Ethernet Module	CPU module with built-in Ethernet port. Executes the ladder logic program and handles network communications.
PLC Rack (DirectLOGIC 205) – Slot 2	Slot 2	0	8	0	0	0	D2-08ND3	8-Point Discrete Input Module	8-channel 24VDC discrete (digital) input module. Accepts sink/source wiring. Used for reading on/off signals such as pushbuttons, limit switches, and proximity sensors.
PLC Rack (DirectLOGIC 205) – Slot 3	Slot 3	0	0	0	8	0	F2-08AD-1	8-Channel Analog Current Input Module (4-20 mA)	8-channel current input module. Used for motor current monitoring (CH1+ to VFD VO terminal, scaled 0-10V from VFD).
PLC Rack (DirectLOGIC 205) – Slot 4	Slot 4	0	0	0	4	0	F2-04RTD	4-Channel RTD Input Module	4-channel RTD (Resistance Temperature Detector) input module. Compatible with PT100/PT1000 and other RTD types. Used for precision temperature measurement.
PLC Rack (DirectLOGIC 205) – Slot 5	Slot 5	8	0	0	0	0	D2-08TR	8-Point Relay Output Module	8-channel relay (dry contact) output module. Common to PSU -V (sinking mode). Drives indicator LEDs (Y0-Y4) and VFD digital inputs (Y5=RST, Y6=FOR, Y7=REV).
PLC Rack (DirectLOGIC 205) – Slot 6	Slot 6	0	0	8	0	0	F2-08DA-2	8-Channel Analog Voltage Output Module (0-10 V)	8-channel voltage output module. CH1 (+V1) provides VFD speed reference to VI terminal (0-10V scaled to frequency).
PLC Rack (DirectLOGIC 205) – Slot 7	Slot 7	—	—	—	—	—	D2-Fill	Fill Panel	Blank fill panel. No I/O. Slot reserved for future expansion.
PLC Rack (DirectLOGIC 205) – Slot 8	Slot 8	—	—	—	—	—	D2-Fill	Fill Panel	Blank fill panel. No I/O. Slot reserved for future expansion.
PLC Rack (DirectLOGIC 205) – Slot 9	Slot 9	—	—	—	—	—	D2-Fill	Fill Panel	Blank fill panel. No I/O. Slot reserved for future expansion.

Sheet: DirectLOGIC CPU (H2-DM1E)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
Power and communication interface	Backplane Connector (Left Edge)	0	0	0	0	1	D2-09B-1 Base Slot 1 Connector	Provides power and backplane communication	–
Ground reference	Backplane GND	0	0	0	0	1	D2-09B-1 Base Ground Bus	Common ground for logic power	–
HMI communication port	CPU Comm Port	0	0	0	0	0	HMI EA1-T4CL COM Port	4" C-More Micro-Graphic panel (operator interface)	CPU communication port links to the C-More HMI (see HMI sheet). Match baud, parity, and stop bits on both ends. Carries operator commands to the PLC and status/data back to the panel.
	Total Points	0	0	0	0	2			

Sheet: Output Module (D2-08TA)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
Output Common (Hot)	C	1	0	0	0	1	120 V Hot Bus	ZipLink Terminal Block (ZL-RTB20-OUT). 1A inline slow blow fuse	120 V hot feed shared by all outputs
Output 0	OUT0	1	0	0	0	0	A1	TX Contactor (GH15-BN)	Drives TX coil via ZipLink
Output 1	OUT1	1	0	0	0	0	A1	BUS Contactor (GH15-BN)	Drives BUS coil via ZipLink
Output 2	OUT2	1	0	0	0	0	A1	F1 Contactor (GH15-BN)	Drives F1 coil via ZipLink
Output 3	OUT3	1	0	0	0	0	95	Thermal Overload Relay (RTD32-180)	Feeds NC contact 95 → 96 → F2 coil
Module Power Feed	Backplane	0	0	0	0	1	Slot 3 Connector	PLC base backplane (120 VAC)	No external wiring
	Total Points	5	0	0	0	2			

Sheet: HMI (EA1-T4CL)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
HMI Serial Comm Port	COM Port	0	0	0	0	0	Serial Port	H2-DM1E	Serial link between PLC CPU and C-More HMI. Configure PLC and HMI ports for matching baud, parity, stop bits
HMI Power +V	+VDC	0	0	0	0	1	+V	Mean Well NDR-480-24	HMI 24 VDC input powered from Mean Well NDR-480-24 +V rail
HMI Power 0 V	0VDC	0	0	0	0	1	0V	Mean Well NDR-480-24	0 VDC return to Mean Well NDR-480-24 -V rail (system DC common)
HMI Ground	Ground	0	0	0	0	1	Ground	Ground	
	Total Points	0	0	0	0	3			

Sheet: Output Module (F2-02DAS-2)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
Analog Common (CH1)	CH1-V	0	0	1	0	0	CM	VFD (GS1-20P2) through Ziplink ZL-RTB20	0V1 and CH1-V are internally shorted
Analog Output CH1	CH1+V	0	0	1	0	0	AL	VFD (GS1-20P2) though Ziplink ZL-RTB20	0–10 V speed command to VFD
Module Logic +24 V	+V1	0	0	0	0	1	VDC +24V	PLC Base (D2-09B-1) through ZipLink ZL-RTB20	Powers analog output circuitry, also fused with 1A fast blow fuse
Module Logic 0 V	0V1	0	0	0	0	1	VDC – 0V	PLC Base (D2-09B-1) through ZipLink ZL-RTB20	Common return for module logic
Slot Power Feed	Backplane	0	0	0	0	1	PLC Base Backplane	Internal slot power	No external wiring
	Total Points	0	0	2	0	3			

Sheet: Input Module (D2-08ND3)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
D2-08ND3 X0	X0	0	1	0	0	0	Forward Pushbutton	Forward button S2PR-P3G with Contact block SA-CA with LED module SA-LDG	Shared button +24V -> NO contact -> X0.
D2-08ND3 X1	X1	0	1	0	0	0	Reverse Pushbutton	Reverse button S2PR-P3Y with Contact block SA-CA with LED module SA-LDY	Shared button +24V -> NO contact -> X1.
D2-08ND3 X2	X2	0	1	0	0	0	Run 1 / Deadman 1 Pushbutton	Run 1 button S2PR-P3B with Contact block SA-CA with LED module SA-LDB	Shared button +24V -> NO contact -> X2.
D2-08ND3 X3	X3	0	1	0	0	0	Run 2 / Deadman 2 Pushbutton	Run 2 button S2PR-P3B with Contact block SA-CA with LED module SA-LDB	Shared button +24V -> NO contact -> X3.
D2-08ND3 X4	X4	0	1	0	0	0	E-Stop Button Monitor	E-stop button E22B1 with NC contact block, wired in series with Mitsubishi SD-N35 contactor coil	PSU +V -> E22B1 NC -> SD-N35 A1 (X4 tapped between E22B1 and A1). SD-N35 A2 -> PSU -V.
D2-08ND3 X5	X5	0	1	0	0	0	VFD Fault Relay	VFD fault output Huanyang HY02D211B-T internal SPDT relay (FA/FC/FB), programmed as fault indication via PD052=02	PSU +V -> VFD FA. VFD FC -> X5. FB unused.
D2-08ND3 X6	X6	0	1	0	0	0	Reserved - Reset Pushbutton (planned)	Reset button S2PR-P3R with Contact block SA-CA with LED module SA-LDR	Unwired. Planned: shared button +24V -> NO contact -> X6 (same topology as X0-X3).
D2-08ND3 X7	X7	0	1	0	0	0	Spare	Unused	Unwired.
D2-08ND3 Input Common (C)	C	0	0	0	0	1	Mean Well NDR-480-24 -V	DC power supply 0V return rail. Both top-row and bottom-row C terminals on the D2-08ND3 are internally tied.	C -> PSU -V. Module wired in sourcing mode. Shared +24V button supply is fed from PSU +V through a 1 A fast-blow fuse (EURO S10H-5H holder) into the EURO S4LH distribution terminal, which provides the common +24V to X0/X1/X2/X3 (and X6 when wired).
Total Points		0	5	0	0	1			

Sheet: Analog Input (F2-08AD-1)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
F2-08AD-1 +24V (Module Power)	+24V	0	0	0	0	1	Mean Well NDR-480-24 +V	DC power supply +24V rail	PSU +V -> Module +24V terminal directly. No fuse on this branch.
F2-08AD-1 0V (Module Power Return)	0V	0	0	0	0	1	Mean Well NDR-480-24 -V	DC power supply 0V return rail. Shared with VFD DCM and ACM (all tied to PSU -V).	PSU -V -> Module 0V terminal. Also serves as the common reference for channel inputs.
F2-08AD-1 CH1+ (Channel 1 Signal)	CH1+	0	0	0	1	0	VFD Motor Current Monitor	VFD analog output Huanyang HY02D211B-T VO terminal, configured for output current via PD054=1 (0-10V proportional to motor current)	VFD VO -> Module CH1+ terminal. Signal return reference via PSU -V (VFD ACM also tied to PSU -V).
F2-08AD-1 CH2	CH2	0	0	0	1	0	Spare	Unused channel	Unwired.
F2-08AD-1 CH3	CH3	0	0	0	1	0	Spare	Unused channel	Unwired.
F2-08AD-1 CH4	CH4	0	0	0	1	0	Spare	Unused channel	Unwired.
F2-08AD-1 CH5	CH5	0	0	0	1	0	Spare	Unused channel	Unwired.
F2-08AD-1 CH6	CH6	0	0	0	1	0	Spare	Unused channel	Unwired.
F2-08AD-1 CH7	CH7	0	0	0	1	0	Spare	Unused channel	Unwired.
F2-08AD-1 CH8	CH8	0	0	0	1	0	Spare	Unused channel	Unwired.
Total Points		0	0	0	1	2			

Sheet: RTD Input (F2-04RTD)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
F2-04RTD +24V (Module Power)	+24V	0	0	0	0	1	Mean Well NDR-480-24 +V	DC power supply +24V rail	PSU +V -> Module +24V terminal. Module installed in rack but no channels currently used in this build.
F2-04RTD 0V (Module Power Return)	0V	0	0	0	0	1	Mean Well NDR-480-24 -V	DC power supply 0V return rail	PSU -V -> Module 0V terminal.
F2-04RTD CH1	CH1	0	0	0	1	0	Spare	Unused RTD input channel	Unwired. Available for future RTD temperature sensor.
F2-04RTD CH2	CH2	0	0	0	1	0	Spare	Unused RTD input channel	Unwired.
F2-04RTD CH3	CH3	0	0	0	1	0	Spare	Unused RTD input channel	Unwired.
F2-04RTD CH4	CH4	0	0	0	1	0	Spare	Unused RTD input channel	Unwired.
Total Points		0	0	0	0	2			

Sheet: Relay Output (D2-08TR)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
D2-08TR Y0	Y0	1	0	0	0	0	Forward Indicator LED	LED module SA-LDG (green, 12-24 VDC) integrated in Forward pushbutton S2PR-P3G (assembly also drives X0)	PSU +V -> SA-LDG (+). SA-LDG (-) -> Y0 contact. Relay sinks Y0 to -V when energized, lighting the LED.
D2-08TR Y1	Y1	1	0	0	0	0	Reverse Indicator LED	LED module SA-LDY (yellow, 12-24 VDC) integrated in Reverse pushbutton S2PR-P3Y (assembly also drives X1)	PSU +V -> SA-LDY (+). SA-LDY (-) -> Y1 contact. Relay sinks Y1 to -V when energized.
D2-08TR Y2	Y2	1	0	0	0	0	Run 1 / Deadman 1 Indicator LED	LED module SA-LDB (blue, 12-24 VDC) integrated in Run 1 pushbutton S2PR-P3B #1 (assembly also drives X2)	PSU +V -> SA-LDB (+). SA-LDB (-) -> Y2 contact.
D2-08TR Y3	Y3	1	0	0	0	0	Run 2 / Deadman 2 Indicator LED	LED module SA-LDB (blue, 12-24 VDC) integrated in Run 2 pushbutton S2PR-P3B #2 (assembly also drives X3)	PSU +V -> SA-LDB (+). SA-LDB (-) -> Y3 contact.
D2-08TR Y4	Y4	1	0	0	0	0	Error / Fault Indicator	Wamco model 525 panel-mount LED indicator (28 VDC, 0.6 W)	PSU +V -> Wamco 525 (+). Wamco 525 (-) -> Y4 contact. 28V-rated indicator slightly under-driven at 24V (acceptable, ~21 mA draw).
D2-08TR Y5	Y5	1	0	0	0	0	VFD Fault Reset	Huanyang HY02D211B-T VFD RST terminal (multi-input, sinking/NPN mode)	Y5 -> VFD RST. When Y5 energized, relay closes Y5 to common (PSU -V), pulling RST input to DCM. ~500 ms pulse during PLC unjam routine.
D2-08TR Y6	Y6	1	0	0	0	0	VFD Forward Run	Huanyang HY02D211B-T VFD FOR terminal (multi-input, sinking/NPN mode)	Y6 -> VFD FOR. When Y6 energized, pulls FOR to DCM. Forward shredding.
D2-08TR Y7	Y7	1	0	0	0	0	VFD Reverse Run	Huanyang HY02D211B-T VFD REV terminal (multi-input, sinking/NPN mode)	Y7 -> VFD REV. When Y7 energized, pulls REV to DCM. Used during PLC unjam routine.
D2-08TR Relay Common (C)	C	0	0	0	0	1	Mean Well NDR-480-24 -V	DC power supply 0V return rail	C -> PSU -V. Sinking mode: when any Y0-Y7 closes, that terminal connects to -V.
Total Points		8	0	0	0	1			

Sheet: Analog Output (F2-08DA-2)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
F2-08DA-2 +24V (Module Power)	+24V	0	0	0	0	1	Mean Well NDR-480-24 +V	DC power supply +24V rail	PSU +V -> Module +24V terminal directly. No fuse on this branch.
F2-08DA-2 0V (Module Power Return / Analog Common)	0V	0	0	0	0	1	Mean Well NDR-480-24 -V	DC power supply 0V return rail. Single 0V terminal serves both as power return and analog signal reference for all 8 channels.	PSU -V -> Module 0V terminal. VFD ACM is also tied to PSU -V, so the analog signal return is shared through the common 0V rail.
F2-08DA-2 +V1 (Channel 1 Signal)	+V1	0	0	1	0	0	VFD Analog Speed Reference	Huanyang HY02D211B-T VFD VI terminal (analog voltage input, 0-10 V), used as frequency/RPM reference. Requires PD002=1.	+V1 -> VFD VI. 0-10 V signal scaled to commanded frequency: 0 V = 0 Hz, 10 V = max frequency.
F2-08DA-2 +V2	+V2	0	0	1	0	0	Spare	Unused channel	Unwired.
F2-08DA-2 +V3	+V3	0	0	1	0	0	Spare	Unused channel	Unwired.
F2-08DA-2 +V4	+V4	0	0	1	0	0	Spare	Unused channel	Unwired.
F2-08DA-2 +V5	+V5	0	0	1	0	0	Spare	Unused channel	Unwired.
F2-08DA-2 +V6	+V6	0	0	1	0	0	Spare	Unused channel	Unwired.
F2-08DA-2 +V7	+V7	0	0	1	0	0	Spare	Unused channel	Unwired.
F2-08DA-2 +V8	+V8	0	0	1	0	0	Spare	Unused channel	Unwired.
Total Points		0	0	1	0	2			

Sheet: VFD (Huanyang HY02D211B-T)

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
VFD R (AC Line Hot Input)	R	0	0	0	0	1	AC Mains Hot via main disconnect and 30 A breaker	120 VAC line conductor from wall outlet, routed through panel main disconnect and a 30 A C-curve DIN-rail circuit breaker	Wall AC L -> disconnect -> 30 A breaker -> VFD R. Black wire per NFPA 79.
VFD T (AC Neutral Input via Contactor and Breaker)	T	0	0	0	0	1	AC Mains Neutral via Phoenix Contact TMC 8 3C 15A breaker and Mitsubishi SD-N35 contactor	Neutral leg switched through the safety-disconnect contactor. As-built path: Wall N -> Phoenix Contact TMC 8 3C 15A breaker (P/N 2907618) -> SD-N35 T1 -> contactor pole -> SD-N35 L1 -> VFD T.	AS-BUILT: neutral leg is switched by the contactor instead of the hot leg. Note: NFPA 79 / NEC standard practice is to switch the HOT leg. Review for final report. When E-stop is pressed, SD-N35 coil drops -> contact opens -> VFD T loses neutral -> VFD shuts down.
VFD Ground (PE)	Ground	0	0	0	0	1	Panel Ground Bus	Equipment grounding conductor; bonds VFD chassis to panel ground bus	VFD ground stud -> panel ground bus. Green or Green/Yellow wire.
VFD U (Motor Phase 1 Output)	U	0	0	0	0	1	Banana Socket (Blue) -> Motor T1	Panel-mount banana socket (blue, 4 mm) provides quick-disconnect to GE 5KE182BC205B motor T1	VFD U -> blue banana socket -> motor terminal.
VFD V (Motor Phase 2 Output)	V	0	0	0	0	1	Banana Socket (White) -> Motor T2	Panel-mount banana socket (white, 4 mm) provides quick-disconnect to GE 5KE182BC205B motor T2	VFD V -> white banana socket -> motor terminal.
VFD W (Motor Phase 3 Output)	W	0	0	0	0	1	Banana Socket (Red) -> Motor T3	Panel-mount banana socket (red, 4 mm) provides quick-disconnect to GE 5KE182BC205B motor T3	VFD W -> red banana socket -> motor terminal.
VFD P+ (Brake Resistor +)	P+	0	0	0	0	1	Brake Resistor ATO APCS-300R30-AD (+ terminal)	Dynamic braking resistor 300 W, 30 Ohms, aluminum-clad (ATO model APCS-300R30-AD)	VFD P+ -> brake resistor terminal 1.
VFD PR (Brake Resistor Return)	PR	0	0	0	0	1	Brake Resistor ATO APCS-300R30-AD (return terminal)	Dynamic braking resistor (same component as P+ row)	VFD PR -> brake resistor terminal 2.
VFD UPF	UPF	0	1	0	0	0	Unused	Multi-Output 2 (PD051), open-collector optocoupler	Unwired.

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
VFD DRV	DRV	0	1	0	0	0	Unused	Multi-Output 1 (PD050), open-collector optocoupler	Unwired.
VFD DCM (Digital Common, top row)	DCM	0	0	0	0	1	Mean Well NDR-480-24 -V	DC power supply 0V return - shared reference for FOR/REV/RST sinking inputs	VFD DCM (top row) -> PSU -V. PSU -V also feeds D2-08TR relay common.
VFD SPL	SPL	0	1	0	0	0	Unused	Multi-Input 6 - default low-speed select	Unwired.
VFD SPM	SPM	0	1	0	0	0	Unused	Multi-Input 5 - default middle-speed select	Unwired.
VFD SPH	SPH	0	1	0	0	0	Unused	Multi-Input 4 - default high-speed select	Unwired.
VFD RST (Fault Reset Input)	RST	0	1	0	0	0	D2-08TR Y5	PLC relay output (sinking NPN mode) driving fault reset	PLC Y5 -> VFD RST. Pulled to DCM (0V) when Y5 closes.
VFD REV (Reverse Run Input)	REV	0	1	0	0	0	D2-08TR Y7	PLC relay output (sinking NPN mode) driving reverse run command	PLC Y7 -> VFD REV. Pulled to DCM (0V) when Y7 closes.
VFD FOR (Forward Run Input)	FOR	0	1	0	0	0	D2-08TR Y6	PLC relay output (sinking NPN mode) driving forward run command	PLC Y6 -> VFD FOR. Pulled to DCM (0V) when Y6 closes.
VFD ACM (Analog Common, top row)	ACM	0	0	0	0	1	Mean Well NDR-480-24 -V	DC power supply 0V return - shared reference for analog signals	VFD ACM (top row) -> PSU -V. Shared 0V reference for VI (speed in) and VO (current monitor out).
VFD VO (Analog Output, 0-10 V)	VO	0	0	1	0	0	F2-08AD-1 CH1+	PLC analog input module CH1, used as motor current monitor (PD054=1)	VFD VO -> F2-08AD-1 CH1+. 0-10 V scaled to motor output current.
VFD 10V	10V	0	0	1	0	0	Unused	+10 V reference output (for external potentiometer)	Unwired.
VFD FA (Multi-Output 3 Relay NO)	FA	0	1	0	0	0	Mean Well NDR-480-24 +V	DC power supply +V rail	PSU +V -> VFD FA. No fuse on this branch. FA-FC closes when VFD trips on any fault (PD052=02).
VFD FC (Multi-Output 3 Relay Common)	FC	0	1	0	0	0	D2-08ND3 X5	PLC discrete input X5 (jam signal)	VFD FC -> PLC X5. When FA-FC closes, +24V from FA reaches X5 (PLC reads HIGH = jam).
VFD FB (Multi-Output 3 Relay NC)	FB	0	1	0	0	0	Unused	Normally-closed contact - not used in active-high NO configuration	Unwired.
VFD 24V (Onboard 24V Supply)	24V	0	0	0	0	1	Unused	Onboard 24 VDC auxiliary supply, max 200 mA	Unwired. Panel uses external Mean Well NDR-480-24 instead.

Point Description	Origin	DO	DI	AO	AI	Pwr	Destination Address	Destination Description	Notes
VFD DCM (bottom row)	DCM (bottom)	0	0	0	0	1	(same node as top-row DCM)	Internally tied to top-row DCM	Alternate landing point. Top-row DCM is used.
VFD 5V	5V	0	0	1	0	0	Unused	+5 V reference output	Unwired.
VFD ACM (bottom row)	ACM (bottom)	0	0	0	0	1	(same node as top-row ACM)	Internally tied to top-row ACM	Alternate landing point. Top-row ACM is used.
VFD AI (Analog Current Input, 4-20 mA)	AI	0	0	0	1	0	Unused	Analog current input for speed reference. Speed command uses VI voltage input instead.	Unwired.
VFD VI (Analog Voltage Input, 0-10 V)	VI	0	0	0	1	0	F2-08DA-2 +V1	PLC analog output module CH1, used as commanded frequency reference (PD002=1)	F2-08DA-2 +V1 -> VFD VI. 0-10 V scaled to commanded frequency.
VFD RS-	RS-	0	0	0	0	1	Unused	RS-485 Modbus communication negative line	Unwired. Modbus not used.
VFD RS+	RS+	0	0	0	0	1	Unused	RS-485 Modbus communication positive line	Unwired.
Total Points		0	5	1	1	12			